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Stat 153/248: Introduction to Time Series Analysis

Instructor and GSIs

- **Instructor** Liberty Hamilton - liberty.hamilton@berkeley.edu
 - **Office Hours** (Evans 351) - Tuesday 9:30–10:30am
- **GSIs:**
 - Nicholas Liskij - nliskij@berkeley.edu
 - **Office Hours:** TBD
 - Yichen Pan - yichenpan@berkeley.edu
 - **Office Hours:**
- **Readers:** Brandon Xu and Jack Zhang

Important Info

- **Class times:** Tuesdays and Thursdays, 8–9:30am, VLSB 2060
- **Lab times:** Fridays, 9–11am, 11–1pm Evans 330 and 1–3pm, 3–5pm, Evans 342
- **Ed discussion** We have a site on [Ed Discussions](#) where you can post questions and interact with others in the class.

- **Github** The [github site](#) will have homeworks and labs, but these will also be posted to this site.
- **bCourses** [bCourses](#) will be used to keep track of grades.

Course Information

A time series is a sequence of data points collected over time, for example, hourly temperature readings, stock market fluctuations, or brain responses over time for a particular stimulus. Such methods are important for uncovering regular patterns over time, including cycles and trends. In this class we will learn techniques for interpreting and modeling such data. Content will include (a) Multiple linear regression models, (b) Nonlinear regression models, (c) Regularized High-dimensional linear regression models, (d) Variance models and spectral analysis, (e) Lagged regressions and ARIMA models, (f) Recurrent Neural Networks, (g) self-supervised models. This class will use the Python programming language. Multiple applications will be explored, including but not limited to speech processing, neuroscience, astronomy, and epidemiology.

Prerequisites

Undergraduate probability at the level of STAT 134 or DATA 140 or EE 126 is required. Statistics at the level of STAT 133 and STAT 135 are strongly recommended. STAT 135 may be taken concurrently, though some students have found this course challenging without prior completion of STAT 135. Labs, homework, and projects will be completed in the Python language, and familiarity with Python is a prerequisite.

Required Materials

Required Texts: There are no required textbooks for this course. Instead, I will provide lecture notes as well as book chapters as PDFs, some of which come from the following recommended texts:

- Recommended Texts: [Shumway & Stoffer Time Series Analysis and Its Applications](#). 4th or 5th edition. 5th edition is available online for free through Springer Books.

Exams

For STAT 153, there will be two exams: Midterm and Final. The Midterm will be on March 19 in class. The Final exam will be on TBD.

For STAT 248, there will be one exam: Midterm (in class on March 19). Instead of the final exam, there will be a final project (details will be announced later).

Grading

Grades will be based on five homework assignments, one midterm exam, and one final exam (STAT 153) or final project (STAT 248). The grading breakdown is as follows (each homework assignment is worth an equal amount):

- Homeworks: 50%
- Midterm: 20%
- Final Exam (Stat153) or Final Project (Stat248): 30%

Differences between Stat153 and Stat248

This class is crosslisted as both an undergraduate (Stat158) and a graduate (Stat248) class. Each homework assignment will have 1-3 additional questions that only students taking STAT248 need to answer. STAT 153 has a final exam while STAT 248 will have a final project.

Late policy

You have a total of 120 late hours that you can apply to your homework for the entire semester. No points will be awarded for any homework which brings the total late hours to more than 120.

Accommodations for students with disabilities

If you require course accommodations due to a physical, emotional, or learning disability, contact UC Berkeley's Disabled Students' Program (DSP). You must have a Letter of Accommodation on file with UC Berkeley to have accommodations made in the course. Please also contact the Instructor (Liberty Hamilton) as early as possible to discuss these accommodations within the first few weeks of class.

Academic Integrity

You are encouraged to work in small groups on homework problems. However, you must write up solutions on your own, including writing your own code, and you must never read or copy the solutions of other students. Similarly, you may use books or online resources to help solve homework problems, but you must credit all such sources in your writeup and you must never copy

materials verbatim. Any students found to be cheating automatically risks failing the class and being referred to the Office of Student Conduct. In particular, copying solutions, in whole or in part, from other students in the class or any other source without acknowledgement constitutes cheating.

Any test, homework, paper, or report submitted by you and that bears your name is presumed to be your own original work that has not previously been submitted for credit in another course unless you obtain prior written approval to do so from your instructor. If you are unclear about the expectations for completing an assignment or taking a test or examination, be sure to seek clarification from your instructor or GSI beforehand. For additional information on plagiarism and how to avoid it, read the UC Berkeley Library Citation Page, Plagiarism Section.

As a member of the campus community, you are expected to demonstrate integrity in all of your academic endeavors and will be evaluated on your own merits. The consequences of cheating and academic dishonesty—including a formal discipline file, possible loss of future internship, scholarship, or employment opportunities, and denial of admission to graduate school—are simply not worth it. Read more about Berkeley's Honor Code.

Take care of yourself

It is important to take care of your mental and physical health. If you need help or support, there are many resources on campus that can help. Please find them at the [Academic Accommodations Hub](#).

Tentative Schedule

Topics will be available on the [course website](#) as they occur. Below is a plan for the semester (subject to change as we move through):

Date	Topic	Assignments
Jan. 20	Introduction to the class	
Jan. 22	Characteristics of time series data	
Jan. 27	Measures of dependence	
Jan. 29	Measures of dependence (cont'd)	HW1 posted
Feb. 3	Simple Linear Regression	

Feb. 5	Simple Linear Regression (cont'd)	
Feb. 10	Multiple Linear Regression	HW1 due
Feb. 12	Nonlinear Regression	HW2 available
Feb. 17	Smoothing and regularization (ridge, LASSO)	
Feb. 19	Smoothing and regularization (ridge, LASSO - cont'd)	
Feb. 24	Smoothing and regularization (ridge, LASSO)	
Feb. 26	Smoothing and regularization (ridge, LASSO - cont'd)	
Mar. 3	Power spectral analysis	HW2 due
Mar. 5	Power spectral analysis (cont'd)	HW3 available
Mar. 10	Time frequency analysis	
Mar. 12	Time frequency analysis (con't)	HW3 due Mar. 13
Mar. 17	MIDTERM EXAM (in class)	
Mar. 19	AR models	
Mar. 24	SPRING BREAK	
Mar. 26	SPRING BREAK	
Mar. 31	ARIMA Models	HW4 available
Apr. 2	ARIMA models (cont'd)	
Apr. 7	Time-lagged regression	
Apr. 9	Time-lagged regression (cont'd)	
Apr. 14	State space models	
Apr. 16	State space models (cont'd)	HW4 due
Apr. 21	Convolutional Neural Networks	HW5 available
Apr. 23	Convolutional Neural Networks (cont'd)	

Apr. 28	Recurrent Neural Networks	
Apr. 30	Recurrent Neural Networks (cont'd)	
May 5	Self-supervised learning	
May 7	Self-supervised learning (cont'd), wrap up	HW5 due
Date TBD	FINAL EXAM	

References

1. Shumway, R. H., & Stoffer, D. S. (2025). Time Series Analysis and Its Applications: With R Examples. In *Springer Texts in Statistics*. Springer Nature Switzerland. [10.1007/978-3-031-70584-7](https://doi.org/10.1007/978-3-031-70584-7)